

# Root compression becomes scalar in norm

MathlibAnnex.CStarAlgebra.CAR.tendsto\_norm\_compressionError

## theorem

Extends an exact rank-one compression identity on finite stages to a pointwise norm limit on the completed algebra.

## Statement

Let  $A$  be the completed CAR algebra with stage maps  $\iota_n : M_{2^n}(\mathbb{C}) \rightarrow A$  and root state  $\omega(\iota_n(a)) = a_{00}$ . Put  $f_n = \iota_n(E_{00})$ , where  $E_{00}$  is the projection onto the first coordinate. Then, for every fixed  $x \in A$ ,  $\|f_n x f_n - \omega(x) f_n\| \rightarrow 0$  as  $n \rightarrow \infty$ .

## Assumptions

The stages are linked by  $a \mapsto a \otimes I_2$ , and  $A$  is their norm completion. The functional  $\omega$  is the continuous root state. Each  $f_n$  is a self-adjoint projection of norm at most one. The element  $x$  is arbitrary but fixed while  $n$  tends to infinity.

## Conclusion

Writing  $E_n(x) = f_n x f_n - \omega(x) f_n$ , the theorem gives  $\lim_{n \rightarrow \infty} \|E_n(x)\| = 0$  for every  $x \in A$ . The error  $E_n(x)$  is an element of  $A$ ; its norm is the real-valued quantity that converges.

This is pointwise convergence in  $x$ , not convergence in operator norm uniformly over the unit ball. It also does not assert norm convergence of the projections  $f_n$ . The displayed projections have rank one at their defining finite stage; no rank-one assertion in an arbitrary representation of  $A$  is used.

## Proof route

If  $y = \iota_m(a)$  lies in a finite stage, rank-one compression gives  $E_n(y) = 0$  whenever  $n \geq m$ . For arbitrary  $z \in A$ , the projection norm bound and boundedness of  $\omega$  give  $\|E_n(z)\| \leq (1 + \|\omega\|)\|z\|$ , uniformly in  $n$ . Approximate the fixed  $x$  by such a  $y$ , and apply this estimate to  $x - y$ .

### ▼ Proof steps

1. For  $\varepsilon > 0$ , choose  $y = \iota_m(a)$  with  $\|x - y\| < \varepsilon/(\|\omega\| + 2)$ , using density of the stages.
2. For  $n \geq m$ , the exact finite-stage compression identity gives  $E_n(y) = 0$ .
3. Linearity gives  $E_n(x) = E_n(x - y)$ . Hence  $\|E_n(x)\| \leq (1 + \|\omega\|)\|x - y\| < \varepsilon$ .
4. This bound holds for every  $n \geq m$ , which is the required norm limit.

## Main citations

- Exact declaration and proof
- MathlibAnnex.CStarAlgebra.CAR.rootState
- MathlibAnnex.CStarAlgebra.CAR.rootState\_stage
- MathlibAnnex.CStarAlgebra.CAR.rootFlag
- MathlibAnnex.CStarAlgebra.CAR.isStarProjection\_rootFlag
- MathlibAnnex.CStarAlgebra.CAR.compressionError
- MathlibAnnex.CStarAlgebra.CAR.rootFlag\_mul\_ofStage\_mul
- MathlibAnnex.CStarAlgebra.CAR.compressionError\_sub
- MathlibAnnex.CStarAlgebra.CAR.norm\_compressionError\_le
- MathlibAnnex.CStarAlgebra.CAR.compressionError\_ofStage
- MathlibAnnex.CStarAlgebra.CAR.dense\_stageRange

## Lean source signature (exact)

```
theorem tendsto_norm_compressionError (x : Limit) :
  Filter.Tendsto (fun n => ||compressionError n x||) Filter.atTop (nhds 0)
```

| In the source                                           | Mathematical meaning                                                                                                                                                                                   |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>x : Limit</code>                                  | A fixed, arbitrary element $x \in A$ of the completed CAR algebra.                                                                                                                                     |
| <code>compressionError n x</code>                       | The element $E_n(x) = f_n x f_n - \omega(x) f_n \in A$ , where $f_n = \iota_n(E_{00})$ and $\omega(\iota_m(a)) = a_{00}$ .                                                                             |
| <code>fun n =&gt;   compressionError n x  </code>       | The real sequence $n \mapsto \ f_n x f_n - \omega(x) f_n\ $ using the C*-norm of $A$ .                                                                                                                 |
| <code>Filter.atTop; nhds 0</code>                       | Respectively $n \rightarrow \infty$ in the natural-number order and convergence to the real number 0.                                                                                                  |
| <code>Filter.Tendsto (...) Filter.atTop (nhds 0)</code> | For this $x$ and each $\varepsilon > 0$ , there is $N$ such that $n \geq N$ implies $\ E_n(x)\  < \varepsilon$ . The threshold may depend on $x$ ; this is not uniform convergence over the unit ball. |

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### ▼ Exact content identity

Declaration: MathlibAnnex.CStarAlgebra.CAR.tendsto\_norm\_compressionError

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